

The University of Sheffield

DEPARTMENT OF COMPUTER SCIENCE

AUTUMN SEMESTER 2004–2005 **2 HOURS**

NETWORK PERFORMANCE ANALYSIS

Answer the question in Section A and TWO questions from Section B.

All questions carry equal weight. Figures in square brackets indicate the percentage of available marks allocated to each part of a question.

Section A

1. This question concerns queueing models based around birth-death processes, *i.e.* Markov arrivals and Markov departures.

a) Explain what you understand by the term utilization factor. What are the advantages and disadvantages of utilisation factors that are close to one?

[25%]

The utilization factor is a measure of the *efficiency of the system*. It is the *percentage of time* that the servers are in use. A high utilization factor means that a *high proportion of the available resources are being used* which is an advantage. The disadvantage *i*s that for utilization factors which are close to one the queue occupancy is likely to be very high and *queueing delays* will be large resulting in poor quality of service. ***Five for each italicised statement.***

b) Give the utilization factors for the following queues: $M/M/1$, $M/M/1/m$, $M/M/m$ and $M/M/\infty$.

[20%]

Queue	Conditions	
$M/M/1$	$\rho = \frac{\lambda}{\mu}$	Where $\mu$is the service rate and λ is the arrival rate.
$M/M/1/m$	$\rho = (1 - P(m))\frac{\lambda}{\mu}$	Where $\mu$is the service rate, λ is the arrival rate and $P(m)$ is the blocking probability.
$M/M/m/m$	$\rho = \frac{\lambda}{m\mu}$	Where $\mu$is the service rate, λ is the arrival rate and m is the number of servers.
$M/M/\infty$	ρdoesn't really exist	The value doesn't exist because there are infinite servers the probability that any part of the system is in use is $1 - P(0) = 1 - ex$

Five for each queue

c) What do you understand by the term 'blocking probability'? Give the blocking probability for the $M/M/m/m$ and $M/M/1/m$ queues.

[20%]

In queues with finite size (such as $M/M/1/m$) customers that arrive when the queue is full are blocked. The probability of this happening in steady state is known as the blocking probability.

Queue	Blocking probability	
$M/M/1/m$	$P(m) = \frac{\rho^m(1-\rho)}{1-\rho^{m+1}}$	Where $\rho = \frac{\lambda}{\mu}$ is the utilisation factor.
$M/M/m/m$	$B(m,a) = P(m) = \frac{\frac{a^m}{m!}}{\sum_{n=0}^{m}\frac{a^n}{n!}}$	Where $a = \frac{\lambda}{\mu}$ is the offered load.

Ten for statement, five for each blocking probability.

d) Describe what you understand by the *steady state distribution* of a queueing system.

[15%]

The steady state distribution of a queue is the distribution *stationary distribution* that the Markov chain converges to. It occurs when a *dynamic equilibrium* is obtained in the queueing system. A dynamic equilibrium is when change in one direction is balanced (on average) by change in the opposite direction. The steady state distribution is *used for making predictions* about the expected states of the queue.

Five for each italicised statement.

e) For the $M/M/1$, $M/M/m$, $M/M/m/m$ and the $M/M/\infty$ queues give the conditions under which the steady state distribution will exist.

[20%]

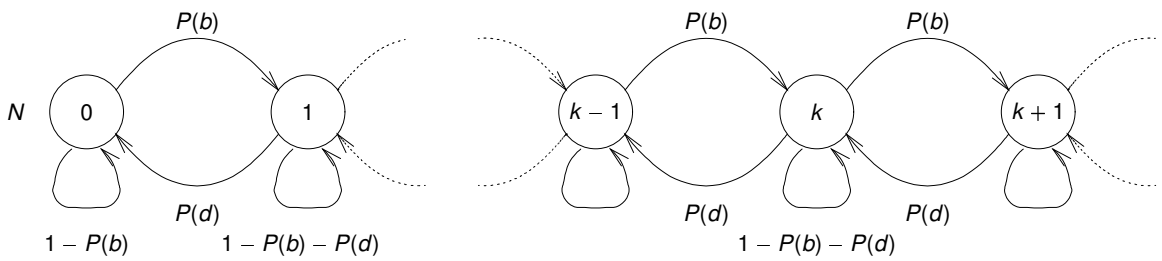
The steady state condition exists for each of these queues when the following conditions are satisfied:

Queue	Conditions	
$M/M/1$	$\lambda < \mu$	The arrival rate is lower than the service rate.
$M/M/m$	$\lambda < m\mu$	The arrival rate is lower than m times the service rate.
$M/M/1/m$	No conditions	The steady state distribution always exists.
$M/M/\infty$	No conditions	The steady state distribution always exists.

Five marks for each queue.

Section B

2. The diagram below gives the state diagram for a discrete time $M/M/1$ queue.

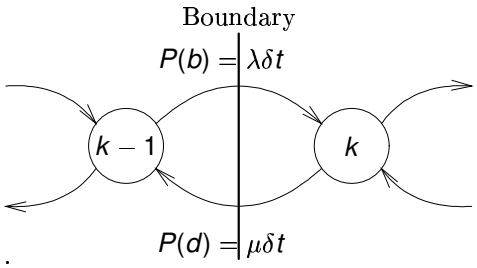


In the system you may take the probability of a birth to be $P(b) = \lambda\delta t$ and the probability of a death to be $P(d) = \mu\delta t$, where δt is the change in time for each time step and $(\mu+\lambda)\,\delta t < 1$.

a) Derive the steady state distribution for this queue in terms of its utilization factor.

[30%]

As a birth-death process the stationary distribution of the queue may be found using detailed balance. We therefore look at the flow across a boundary between two states.



For this flow to balance we must have, at steady state,

$$P(k)\,P(d) = P(k-1)\,P(b)$$

$$P(k)\,\mu\delta t = P(k-1)\,\lambda\delta t$$

$$P(k)\,\mu = P(k-1)\,\lambda$$

$$P(k) = \frac{\lambda}{\mu}P(k-1)$$

Taking the utilisation factor $\rho = \frac{\lambda}{\mu}$ we have

$$P(k) = \rho P(k-1)$$

$$P(k) = \rho^k P(0)$$

Using the normalisation condition to find $P(0)$

$$\sum_{k=0}^{\infty} P(k) = 1$$

$$P(0)\sum_{k=0}^{\infty} \rho^k = 1$$

The infinite series converges for $\rho < 1$ (equivalent to $\mu > \lambda$) giving

$$\sum_{k=0}^{\infty} = \frac{1}{1-\rho}$$

substituting back in we have

$$P(0) = 1 - \rho$$

and finally

$$P(k) = \rho^k (1-\rho)$$

which is recognised as a geometric distribution.

5 for the boundary flow diagram 10 for the correct value in terms of $P(0)$, 10 for the correct derivation of $P(0)$ and 5 for the right answer.

b) Derive the mean of this distribution.

[20%]

The mean has the form

$$\langle k \rangle = \sum_{k=0}^{\infty} k\rho^k (1-\rho)$$

Extracting one $\rho(1-\rho)$ we have

$$\langle k \rangle = \rho(1-\rho) \sum_{k=0}^{\infty} k\rho^{k-1}$$

which is recognised as

$$\langle k \rangle = \rho(1-\rho) \frac{d}{d\rho} \sum_{k=0}^{\infty} \rho^k$$

which can be written

$$\langle k \rangle = \rho(1-\rho) \frac{d(1-\rho)^{-1}}{d\rho}$$

$$\langle k \rangle = \rho(1-\rho) \frac{1}{(1-\rho)^2}$$

$$\langle k \rangle = \frac{\rho}{1-\rho}.$$

***5 for the correct formula for the mean, 10 for the differentiation trick and 5 for the correct answer.*0**

c) Write down an algorithm for computer simulation of this queue. You may use MATLAB notation or pseudocode to express your algorithm.

[30%]

```
lambda = 1;  % birth rate
mu = 2;   % death rate
deltat = 0.01; % time step
k = 0;  % Initial state of queue
T = 100000; % a number of time steps
for i = 1:T
  p = rand(1); % a uniform random number between 0 and 1
  if p < lambda*deltat
    k = k + 1;
  elseif p < (lambda+mu)*deltat & k > 0
    k = k - 1;
  end
end
```

***10 marks for overall structure of code, 5 for random number generation 5 for first if statement and 10 for second if statement (5 for each condition).*0**

d) When designing networks we are normally interested in the steady state behaviour of the system. For simple queues such as the discrete time $M/M/1$ we can compute this analytically. If we cannot compute it analytically we have to resort to simulation. What problems could occur with the simulation you described in part c) which might mean you do not accurately estimate the steady state distribution? You may wish to base your answer around the three different averages: distribution, sample based and time.

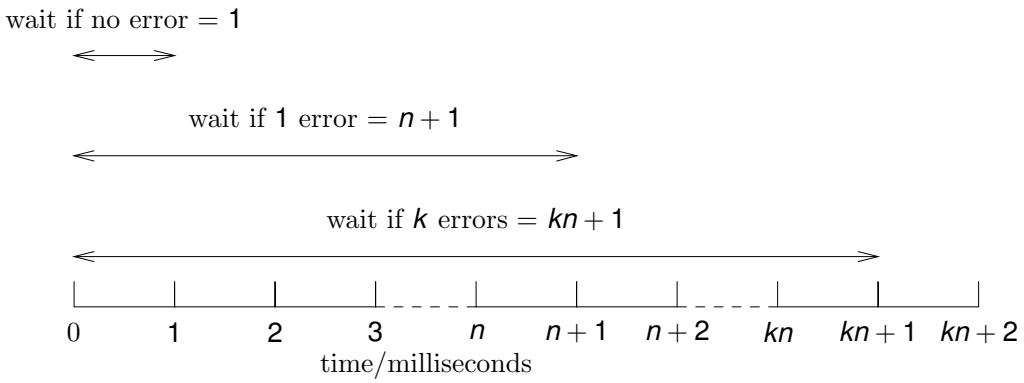
[20%]

Typically in queueing problems we are interested in expectations under steady state distribution. Ideally we want to compute the steady state distribution's average, which we can do for the discrete $M/M/1$ queue. However for more complex queueing systems we may not be able to compute this average in which case we try and approximated it with a sample based average. Typically these approximations involve sampling from a Markov chain. Averaging these values directly leads to a time based average, for a sample based average we need to samples to be *independent*. To estimate the sample based average accurately we do not take every sample from the Markov chain, but we wait between samples.

Finally if the effective state space is large (the state space that the queue is likely to visit) then we may have to wait a long time for enough samples to get an accurate idea of the steady state distribution, it is also difficult to tell how long we need to wait.

10 for averages correctly described. 5 for understanding sample based average vs time dependent and 5 for saying we may not run simulation long enough.

3. Delay analysis of an ARQ System. In an automatic repeat request (ARQ) system packets are transmitted in frames that are 1 millisecond long. An acknowledgement is sent if a packet is received without error. The system waits for a maximum of $n-1$ milliseconds to receive the acknowledgement. Once the acknowledgement is received the next packet in the queue is transmitted. A schematic of this system is shown below.



For the purposes of the analysis below you can take probability of an error in a packet to be p and you may assume that errors occur statistically independently. Finally you may assume that arrivals occur as a Poisson process with a rate λ.

a) Explain why the probability distribution for the time interval between the start of the transmission of a given packet and the end of the previous transmission (the service time distribution) is given by

$$P(X = 1+kn) = (1-p)\,p^k$$

[15%]

The distribution represents the probability that there are k errors followed p^k followed by a successful transmission $(1-p)$. The length of time that this takes will be n for each failure (1 for transmission and $n-1$ waiting for the acknowledgement) plus the length of transmission for a single packet. ***five for k retransmission probability 5 for successful transmission probability and 5 for associated length of time.***

b) What is the mean of this distribution?

[25%]

$$\begin{aligned}\langle X\rangle &= \sum_{k=0}^{\infty}(1+kn)(1-p)\,p^k \\ &= (1-p)\left(\sum_{k=0}^{\infty}p^k + n\sum_{k=0}^{\infty}kp^k\right) \\ &= 1+\frac{np}{1-p}\end{aligned}$$

May also need to compute the sum of kp^k. ***10 for mean formula, 5 for first line, 10 for result.***

c) What is the utilization factor for this queue?

[10%]

The utilization is simply the arrival rate multiplied by the average service time.

$$\rho = \frac{\lambda(1-p)}{1-p+np}$$

and thus

$$1-\rho = \frac{1-\lambda-p(n-1-\lambda)}{1-p+np}$$

d) What is the second moment of the service time distribution?

[30%]

$$\begin{aligned}\left\langle X^2\right\rangle &= \sum_{k=0}^{\infty}(1+kn)^2(1-p)\,p^k \\ &= (1-p)\left(\sum_{k=0}^{\infty}p^k + 2n\sum_{k=0}^{\infty}kp^k + n^2\sum_{k=0}^{\infty}k^2p^k\right) \\ &= 1+\frac{2np}{1-p}+\frac{n^2\left(p+p^2\right)}{(1-p)^2}\end{aligned}$$

10 for each line.

e) Comment on the validity of the independence assumption you have made for the packet errors.

[20%]

In practice, this assumption is probably unrealistic. If an error occurs due to problems in the transmission channel (such as interference) it is likely that several packets in a row will not be received, so the errors do not occur independently. Also the assumption of a fixed probability of packet error may be incorrect, *e.g.* it may vary with load.

You may find the following formula useful in your derivations: $\sum_{k=0}^{\infty}k^2p^k=\frac{p+p^2}{(1-p)^3}$.

4. When simulating networks of queues we have to resort to approximations. In this question we will review some of the issues that arise when performing these approximations.

a) A small system is modelled as two queues. Packets arrive at the first queue as a Poisson stream with exponentially distributed packet lengths. After leaving the first queue the packets enter a second queue. The service time for each packet at both queues is proportional to each packet's length. Explain how the second queue's size, as perceived by an arriving packet, will be related to the arriving packet's length. How will this affect the waiting time for the arriving packet at the second queue?

[25%]

The longer packets will arrive at the second queue to find a shorter queue than the shorter packets. Or in other words queue length at the second queue are anti correlated with packet lengths. This is because, while the longer packet was being served at the first queue there is time for the second queue to empty out. This is analogous to a slow moving truck driver who will generally see an open road in front of him.

10 marks for correctly saying that the queue will be shorter for longer packets (or vice versa). 10 Marks for saying the longer packet will wait less time at the second queue. 5 marks for analogy (given in course text and notes).

b) Explain what you understand by Kleinrock's independence assumption. Given this assumption, what is the delay for a transmission line with a traffic rate of λ and an average transmission time of $\bar{X}$?

[25%]

Kleinrock's independence assumption is an assumption that a packets service time at a node in a network of queues is statistically independent of its service time at any other node in the queueing model. In the approximation you assume each communication link is an $M/M/1$ queue. This means the delay is given by

$$\bar{T} = \frac{1}{\mu_{ij} - \lambda_{ij}} = \frac{\bar{X}}{1-\rho}$$

where $\rho = \lambda\bar{X}$, the utilization factor. ***10 marks for the first sentence or equivalent meaning. 15 marks for the approximation of the communication links as an $M/M/1$ queue*** *and correct formula.*

c) Give conditions under which Kleinrock's independence assumption will yield a reasonable approximation for a network of queues.

[25%]

Kleinrock suggested that when different traffic streams are merged to form a single stream the approximation becomes reasonable. Particularly if that traffic is a large volume from outside the network and is Poisson distributed. The approximation also is better for single path routing algorithms. (Example 3.17 in Bertsekas and Gallagher) ***10 marks for merging of streams. 10 marks for if the additional traffic is Poisson and from outside the network. 5 marks for the single path routing algorithms.*** Note that in practice students also suggested Medium/Medium-Heavy traffic loads, Poisson arrivals and exponential service times and densely connected networks, I gave a maximum of 12 marks for answers which mentioned these issues but failed to mention any of the above.

d) What does Jackson's theorem state and what are its assumptions? What is the average number of packets in a system according to Jackson's theorem?

[25%]

Jackson's theorem states that if packet interarrival times after passing through the first queue are not correlated with their lengths (Kleinrock's independence assumption) and if randomisation is used to divert traffic along the different routes then the average number of packets in the system can be derived as if each queue where $M/M/1$. This gives the number of packets in the system as

$$\bar{N} = \sum_{ij} \frac{\lambda_{ij}}{\lambda_{ij} - \mu_{ij}}$$

where λ_{ij} is the traffic flow rate between node i and j and μ_{ij} is the average service time between node i and node j. ***10 marks for randomisation and 5 for Kleinrock. 10 marks for formula.*****0**

END OF QUESTION PAPER